Exp no: 2

23/7/26

STUDY OF NETWORK CABLES

AIM:

Study of different types of network cables

a) Understand different types of network cables,

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial cable

Cable type	Category	maximum data Transmission	Advantages / Disadvantages	Applicatio use
UTP	Category 3	10 bps	Adv: * Cheaper cost * Easy to install Disadv: * More prone to (EMI) * Electromagnetic	10Base-T ethernet
	Category 5	Upto 100 mbps		Fast ethernet,
	Category 5e	1 Gbps		Fast ethernet
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded * Faster than UTP Disadv: * Expensive * greater installation effects.	Gigabit Ethernet 10 G Ethernet (55 m)
SSTP	Category 7	10 Gbps		Gigabit eth 10 G ethernet (100 m)

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Adv: * High bandwidth * Low loss bandwidth * Versatile disadv: * Expensive * Greater	speed of signal is 500 m
fibre optics cable	single mode multi mode	100 Gbps	Adv: * High speed * High bandwidth * Long distance disadv: * Expensive * Requires	Maximum distance of fibre optic cable is around 100 metres

b) Make your own ethernet cross-over cable/straight cable,

Tools and parts needed,

* Ethernet cabling. CAT5e is certified for gigabit support.
* Two RJ45 plugs
* optional two plug shields

RESULTS:

Thus different types of network cables are studied successfully.